

DAYANANDA SAGAR UNIVERSITY

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**SCHOOL OF
ENGINEERING**

**Bachelor of Technology
in
COMPUTER SCIENCE AND TECHNOLOGY**

**Full Stack Development
Project Report**

“DSU WEBSITE”

BY

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**DEPARTMENT OF COMPUTER SCIENCE & TECHNOLOGY,
SCHOOL OF ENGINEERING
DAYANANDA SAGAR UNIVERSITY,
(2025-2026)**

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CERTIFICATE

This is to certify that the Full Stack Development Mini Project work titled “**DSU WEBSITE**” is carried out by **KAPARTHI SKANDHA (ENG24CT0046), GIRIDHAR BM (ENG24CT0040), POORNIMA YK(ENG24CT0067), RASHTRI(ENG24CT0055), YASHWANTH(ENG24CT0063)**, bonafide students of Third semester of Bachelor of Technology in Computer Science and Technology at the School of Engineering, Dayananda Sagar University, Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Technology, during the year **2025-2026**.

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DECLARATION

We,SKANDHA(ENG24CT0046),GIRIDHAR BM(ENG24CT0040),POORNIMA YK (ENG24T0067), RASTRITH(ENG24CT0055), YASHWANTH(ENG24CT0063), are students of Third semester B. Tech in Computer Science and Technology, at School of Engineering, Dayananda Sagar University, hereby declare that the Mini Project titled “Title” has been carried out by us and submitted in partial fulfilment for the award of degree in Bachelor of Technology in Computer Science and Technology during the academic year 2025-2026.

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We are also grateful to our family and friends who provided us with every requirement throughout the course.

We would like to thank one and all who directly or indirectly helped us in the mini Project work.

Abstract

This page will contain a concise and summary of the mini project work. The entire abstract should not be more than 200 words. Start by clearly defining the purpose of your project. It can include some brief context on the social or academic relevance of your topic, but don't go into detailed background information. After identifying the problem, state in 1-2 sentences what's already known about the problem, what previous research has been done or shown state the objective of your research. Use verbs like investigate, test, analyze or evaluate to describe exactly what you set out to do. Next, indicate the research methods that would be used in the project.

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1.INTRODUCTION

The development of a college website as part of the Full Stack Development (FSD) project aims to create a comprehensive, user-friendly, and interactive digital platform that enhances communication, accessibility, and information sharing within the academic environment. In today's digital era, educational institutions rely heavily on web-based systems to provide seamless access to essential services, announcements, academic resources, and administrative information. This project focuses on building a responsive and dynamic college website that caters to the needs of students, faculty, and visitors by providing an organized and intuitive interface.

The website integrates core FSD concepts such as front-end design, back-end development, database connectivity, and API interactions to deliver a complete functional system. The front end emphasizes simplicity, responsiveness, and modern UI practices, ensuring that users can navigate through pages like About Us, Departments, Courses, Admissions, Events, and Contact Information with ease. The back end is designed to securely manage data, support authentication modules, and enable smooth, real-time content updates by administrators.

A key goal of this project is to provide a centralized hub where users can conveniently access academic calendars, notices, faculty details, and campus-related information. Additionally, the platform is built to be scalable, allowing future enhancements such as chatbot support, student login portals, online application forms, and integration of learning resources.

Overall, this project demonstrates the application of full-stack technologies to solve real-world institutional needs while delivering a polished and reliable web solution for a modern college environment.

2.CHALLENGES FACED

During the development of the college website in this Full Stack Development project, several challenges were encountered across different stages of the workflow. One of the major challenges was clearly understanding the functional requirements and deciding how to organize a large amount of academic and administrative information in a structured and user-friendly way. Designing a clean, modern, and responsive user interface required multiple iterations, especially to ensure that the website worked smoothly on various screen sizes, including mobile devices.

Integrating the front end with the backend posed its own difficulties. Handling API requests, managing asynchronous operations, and ensuring secure data transfer required careful planning. Issues such as CORS errors, incorrect routing paths, and API failures needed repeated debugging. Creating a well-structured database was another important challenge, especially while ensuring data consistency, normalization, and efficient retrieval.

Implementing authentication, form validation, and admin-level access added complexity to the project. Ensuring the security of login systems, protecting user data, and managing sessions required additional research and testing. Performance optimization was also challenging, particularly in reducing page load time, compressing media files, and resolving UI lag during rendering.

Deploying the project introduced new challenges related to server configuration, environment variables, hosting limitations, and ensuring backend functionality worked properly on the live server. Continuous testing, fixing broken links, and checking browser compatibility took significant time.

Despite these challenges, the project helped develop problem-solving skills, improved technical understanding, and strengthened practical full stack development experience.

3.TOOLS USED

HTML5 – For structuring the website pages.

CSS3 – For styling, layout design, animations, and responsiveness.

JavaScript – To add interactivity, dynamic elements, and validation.

Bootstrap / Tailwind CSS (whichever you used) – For responsive design and ready-made UI components.

React / Angular / Vue (optional, based on your project) – To build reusable components and manage user interface efficiently.

2. Back-End Tools

Node.js – For server-side scripting and backend logic.

Express.js – To create REST APIs and manage routing.

Java / Python / PHP (if used as backend) – For handling business logic.

3. Database Tools

MySQL / MongoDB / PostgreSQL – For storing student data, notices, faculty details, and forms.

phpMyAdmin / Mongo Compass – For database management and queries.

4. Development & Version Control Tools

VS Code – Main code editor for writing and debugging code.

Git & GitHub – For version control, collaboration, and code backup.

Postman – For testing APIs and backend endpoints.

5. Deployment Tools

Vercel / Netlify / GitHub Pages – For deploying the frontend.

Render / Railway / Heroku – For hosting backend services.

Cpanel / Domain Hosting (if used) – For DNS and server configuration.

6. Additional Utilities

Figma / Canva – For UI/ design mockups.

Chrome DevTools – For debugging front-end issues.

4.PROJECT OUTCOMES

The development of the college website as part of the Full Stack Development project resulted in several meaningful outcomes, both technically and academically. One of the primary outcomes was the successful creation of a fully functional, responsive, and user-friendly platform that provides essential information about the institution, including departments, faculty, courses, events, and contact details. The project demonstrated the ability to integrate front-end and back-end technologies effectively, resulting in a seamless browsing experience for users.

Through this project, significant technical skills were strengthened. Students gained hands-on experience with HTML, CSS, JavaScript, and frameworks, as well as backend technologies such as Node.js, Express, and database systems. The project improved the understanding of API development, routing, server handling, and secure data management. Creating dynamic pages, validating forms, and implementing authentication contributed to deeper knowledge of full stack development concepts.

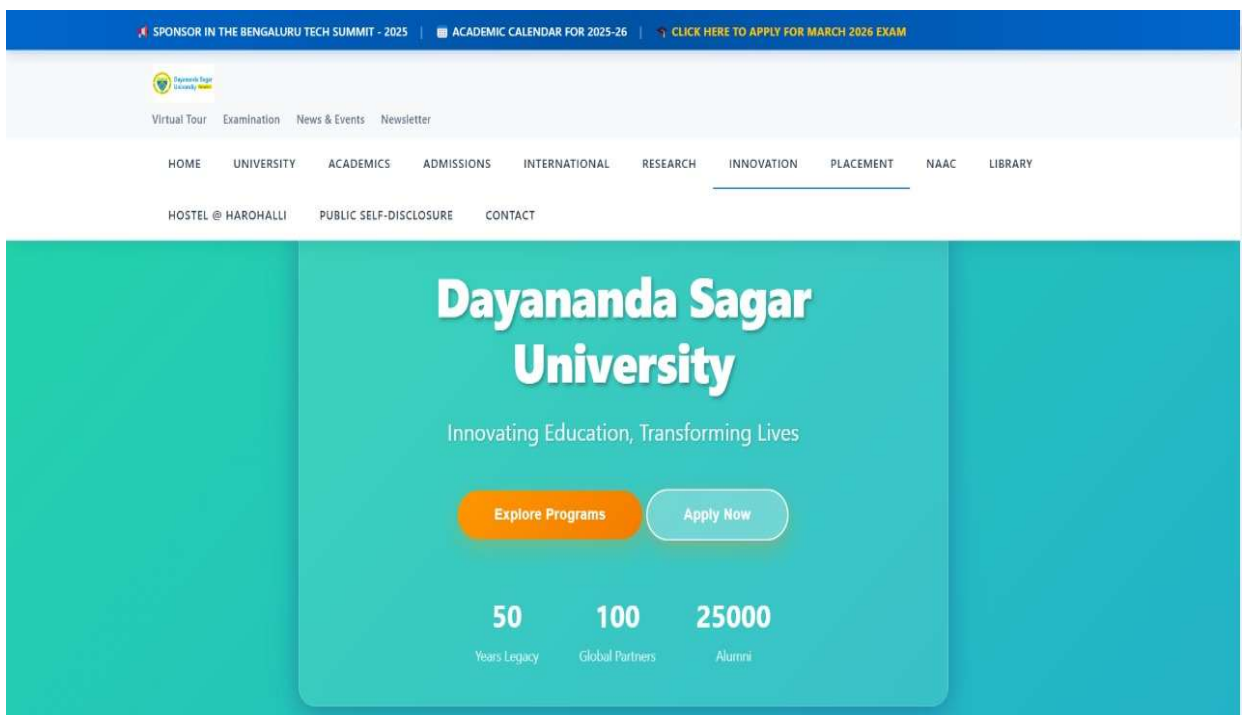
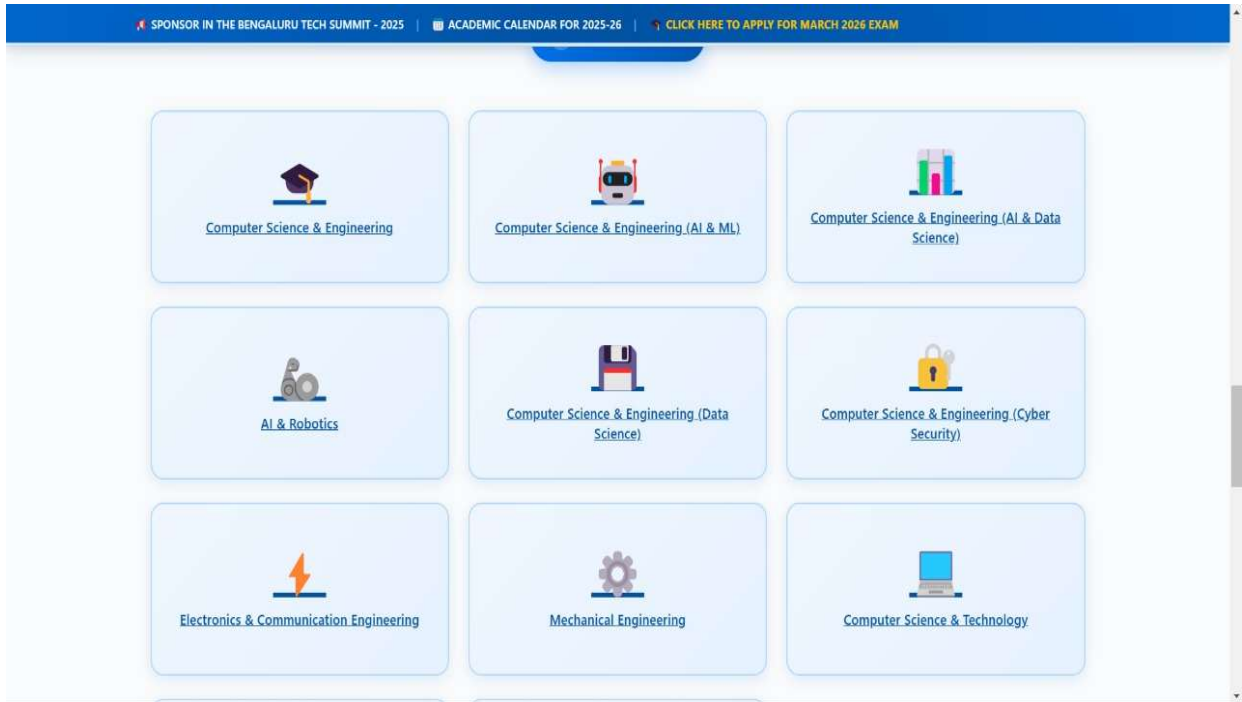
Another important outcome was the enhancement of UI/UX design skills. The website's interface was designed with a focus on simplicity, accessibility, and modern design principles. This helped in developing an eye for layout, color combinations, typography, and overall user experience.

In addition, the project enhanced problem-solving abilities by tackling real-world challenges such as debugging, performance optimization, hosting issues, and cross-browser behavior. It also strengthened teamwork, documentation, and version control practices through tools like Git and GitHub.

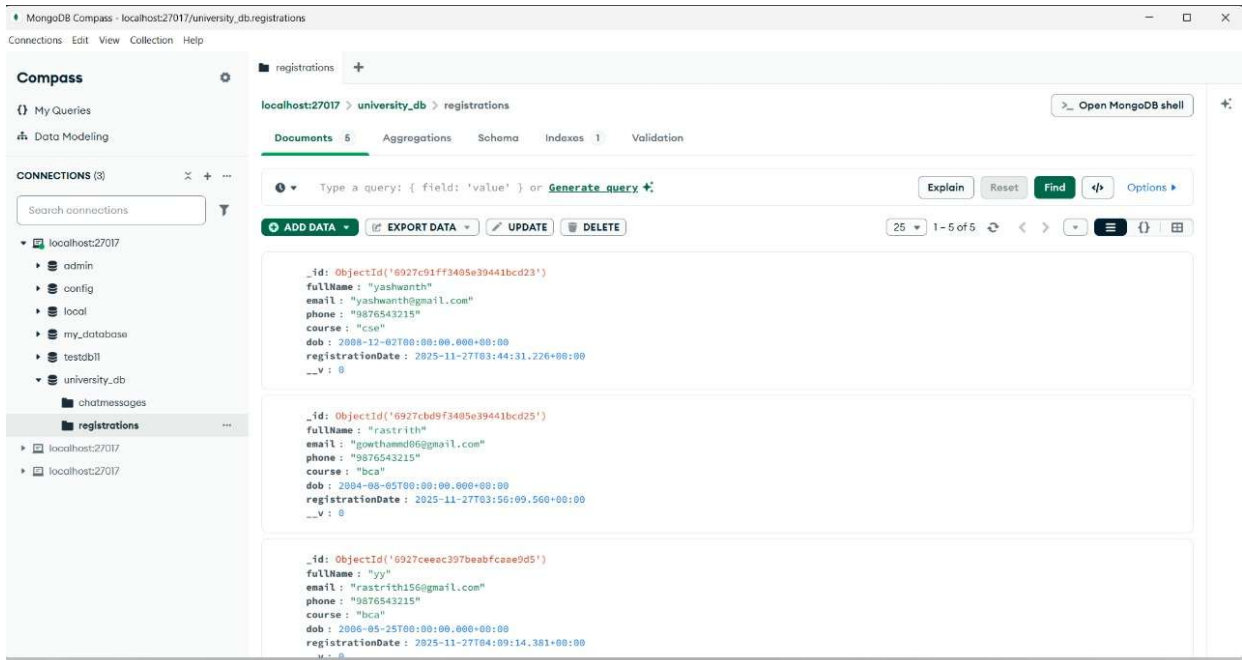
Overall, this project provided a complete practical understanding of how real-world web applications are planned, developed, tested, and deployed, making it a valuable learning experience in full stack development.

5.RESULT

5.1-HTML



5.2-MONGODB



6.CONCULSION

The development of the college website as a full stack project provided a comprehensive and practical learning experience, allowing the integration of both front-end and back-end technologies to build a functional, user-friendly, and dynamic web platform. Throughout the project, various stages of the software development lifecycle were followed, including requirement analysis, design, implementation, testing, and deployment. This structured approach ensured that the final product met the expected objectives and delivered an efficient solution for students, faculty, and administrators.

On the front end, technologies such as HTML, CSS, and JavaScript were utilized to design an intuitive and responsive user interface, ensuring seamless navigation and accessibility across devices. The back end was implemented using robust technologies, enabling secure data management, smooth server-side operations, and efficient communication with the database. By connecting the client and server sides, the project successfully demonstrated the practical application of API integration, authentication mechanisms, and data handling.

Additionally, the project enhanced problem-solving skills by addressing real-world challenges such as optimizing performance, ensuring security, and improving user experience. The testing phase validated the system's reliability and functionality, ensuring a stable and error-free website. Overall, this full stack development project not only achieved its goal of creating a fully functional college website but also strengthened technical expertise and teamwork abilities. It served as a valuable opportunity to explore modern web development practices and laid a strong foundation for future professional projects and learning advancements.

7.GITHUB LINKS

Github repo: <https://rastrith156.github.io/website/>

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GIRIDHAR.B.M	ENG24CT0040	Backend API development, data base
KAPARTHI SKANDHA	ENG24CT0046	Frontend development
POORNIMA.Y.K	ENG24CT0067	AI chat bot integration
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